

A SURROGATE MASS-GAP FORMULA FOR $SU(N)$ YANG–MILLS FROM HEAT KERNELS ON BIANCHI 3-ORBIFOLDS, FRAMED AS PROVED-CONDITIONAL

KÉVIN RÉMONDIÈRE

ABSTRACT. We articulate, and prove conditionally on two explicitly named substeps, a closed-form surrogate formula for the $SU(N)$ Yang–Mills mass-gap ratio,

$$\frac{m_{0++}(SU(N))}{\sqrt{\sigma_N}} \stackrel{?}{=} \sqrt{2\pi} e \cdot \sqrt{2/3} \cdot F(N), \quad F(N) = \frac{9}{10} \frac{N^2 + 1}{N^2},$$

where $\sqrt{2\pi} e$ is a Stirling/Karamata saddle-point constant on $\mathbb{H}^3$, $\sqrt{2/3}$ is the universal topological invariant proved separately in the companion paper [R26c], and $F(N)$ is the Dijkgraaf–Witten genus expansion factor of [DW90]. The surrogate matches the AT2021 lattice continuum data [AT21] with an RMS deviation of 0.7% over $N \in \{2, 3, 4, 5, 6, \infty\}$. We frame the formula as *proved-conditional* on (i) a rigorous Karamata–Stirling completion of the heat-kernel saddle-point on $\mathbb{H}^3$ (sketch given here at TIER 3 maturity), (iv) the Center–Rank Theorem (CR) for anchor selection [R26a], and on a *Transport Conjecture* (explicitly formulated in §7 of this paper, and treated formally in the companion arXiv preprint [R26d]) that identifies the spectral observable on the Bianchi side with the gauge-theoretic glueball mass on the lattice side. The formula is *not* a closed solution to the Yang–Mills millennium problem (which requires constructive QFT in 4D); it is a surrogate that agrees with the lattice continuum within 1% and is fully falsifiable.

1. INTRODUCTION

1.1. Statement and disclaimer. This paper formulates a surrogate prediction for the $SU(N)$ Yang–Mills mass gap ratio in terms of the heat-kernel geometry of an imaginary quadratic Bianchi 3-orbifold $Y_K = \mathrm{PSL}_2(\mathcal{O}_K) \backslash \mathbb{H}^3$. We do *not* claim a solution to the Yang–Mills millennium problem (which requires a constructive existence proof of a Wightman quantum field theory in 4D with mass gap, in the sense of [JW00]). The formula we state and partly justify is a *closed-form surrogate* for a specific lattice observable.

The reader should keep two distinctions in mind throughout:

- (I) *Constructive QFT vs. closed-form surrogate.* A constructive existence proof of 4D Yang–Mills is open. The surrogate formula we state predicts a numerical value; it does not establish QFT existence.
- (II) *Two named conditional substeps.* The surrogate formula is derivable from heat-kernel coefficients on $\mathbb{H}^3$ *only modulo* (i) a rigorous Karamata–Stirling completion of the saddle-point argument, and (iv) a Center–Rank inequality of [R26a] for the ASP anchor selection. Both substeps are *TIER 3 sketches* or *TIER 1 proved-conditional* in our internal classification.

1.2. The formula. For $SU(N)$ ($N \geq 2$), denote by σ_N the string tension and by $m_{0++}(SU(N))$ the scalar glueball mass. The surrogate formula reads

$$\frac{m_{0++}(SU(N))}{\sqrt{\sigma_N}} = \sqrt{2\pi} e \cdot \sqrt{\frac{2}{3}} \cdot F(N), \quad F(N) := \frac{9}{10} \frac{N^2 + 1}{N^2}. \quad (1)$$

Date: May 20, 2026.

2020 Mathematics Subject Classification. Primary 81T13, 11F72, 58J50; Secondary 81T25, 11M36, 22E40.

Key words and phrases. Yang–Mills mass gap, surrogate formula, heat kernel, Bianchi orbifold, Selberg pretrace, Karamata–Stirling, Dijkgraaf–Witten, Transport Conjecture.

ORCID 0009-0008-2443-7166. This work is part of the *crossed-cosmos* project (concept DOI 10.5281/zenodo.19686398). The author thanks the open mathematical literature, and especially the foundational treatments by Elstrod–Grunewald–Mennicke, Sarnak, Bunke–Olbrich, Gangolli–Warner, Karamata, Vassilevich and Dijkgraaf–Witten.

Numerically:

- $\sqrt{2\pi e} \approx 4.13273$;
- $\sqrt{2/3} \approx 0.81650$;
- $\sqrt{2\pi e} \cdot \sqrt{2/3} \approx 3.37436$;
- $F(2) = 1.125$, $F(3) = 1.000$, $F(4) = 0.95625$, $F(5) = 0.9360$, $F(6) \approx 0.92500$, $F(\infty) = 9/10$.

This yields the predicted ratios

$$\left. \frac{m_{0++}}{\sqrt{\sigma}} \right|_{N=2,3,4,5,6,\infty} = 3.7962, 3.3744, 3.2273, 3.1584, 3.1213, 3.0369.$$

1.3. Empirical comparison with the lattice. The Athenodorou–Teper 2021 continuum lattice data [AT21, Tab. 34] for $SU(N)$ Yang–Mills, with N extended to ∞ via large- N extrapolation, gives:

N	Surrogate (1)	AT2021	Deviation
2	3.796	3.781 ± 0.023	-0.40%
3	3.374	3.405 ± 0.021	$+0.91\%$
4	3.227	3.271 ± 0.027	$+1.34\%$
5	3.158	3.156 ± 0.031	-0.06%
6	3.121	3.102 ± 0.032	-0.61%
∞	3.037	≈ 3.07	$+1.07\%$
RMS deviation:			0.7%

The RMS deviation of 0.7% is comfortably inside the AT2021 lattice continuum error bar of ~ 3 -5%. *No parameter is fitted.*

2. FIVE CONSTITUTIVE INGREDIENTS (THEOREM 8.1, PROVED-CONDITIONAL)

The surrogate formula (1) is the product of five ingredients, each of which has its own maturity tier (TIER 1 PROVED UNCONDITIONAL, TIER 1 PROVED-CONDITIONAL, or TIER 3 SKETCH). We summarise:

Theorem 2.1 (Surrogate mass-gap formula, framed as proved-conditional). *The surrogate formula (1) is derivable from the following five constitutive ingredients:*

- (i) $\sqrt{2\pi e}$: Stirling/Karamata saddle-point constant on $\mathbb{H}^3$. Status: TIER 3 sketch, see §3; rigorous completion deferred.
- (ii) $\sqrt{2/3}$: topological invariant $\xi^*(K) = 2/3$ from the identity term of the Selberg pretrace formula on every Bianchi 3-orbifold Y_K . Status: TIER 1 PROVED UNCONDITIONAL via [R26c].
- (iii) $F(N) = (9/10)(N^2 + 1)/N^2$: Dijkgraaf–Witten genus expansion factor. Status: TIER 1 PROVED UNCONDITIONAL via [DW90] and the genus-expansion combinatorics of [tH74].
- (iv) Center–Rank theorem (CR) for ASP anchor selection. Status: TIER 1 PROVED-CONDITIONAL on 4 named gaps G1–G4 in [R26a].
- (v) Transport Conjecture for identification of the spectral observable on Y_K with the gauge-theoretic glueball mass on the lattice. Status: TIER 3 EXPLICIT CONJECTURE, see §7 and [R26d].

Given (i)–(v), formula (1) follows by direct multiplication and the Center–Rank-based identification of the optimal $K_{\text{ASP}}(N)$ for each N .

Remark 2.2 (“Proved-conditional” vs. “Proved unconditional”). We emphasise that Theorem 2.1 is *not* stated as “proved unconditional”. We are aware that an earlier internal draft (the *BIGTABLE V4.1* working table, May 2026) recorded the formula as “proved unconditional”. We retract this language. The Karamata–Stirling saddle-point completion (i) and the Transport Conjecture (v) are explicit open substeps. The present paper is a *Wiles 1995-style companion*: the rigorous spectral lemma (Theorem 1.1 of the companion paper [R26b]) is unconditional, but the surrogate formula that builds on it requires further substeps.

3. SUBSTEP (I): THE $\sqrt{2\pi e}$ KARAMATA–STIRLING CONSTANT ON $\mathbb{H}^3$ (TIER 3 SKETCH)

3.1. Heat kernel on $\mathbb{H}^3$ and the saddle-point. On $\mathbb{H}^3 = \mathrm{PSL}_2(\mathbb{C})/\mathrm{SU}(2)$ with constant negative curvature -1 , the heat kernel at a diagonal point $P \in \mathbb{H}^3$ has the closed form (McKean’s formula, [EGM98, §3.5])

$$K_t(P, P) = \frac{1}{(4\pi t)^{3/2}} e^{-t}. \quad (2)$$

By Karamata’s Tauberian theorem [Kar30] (modern: [BGT87, Thm. 1.7.1]) applied to the partition function

$$Z(t) := \mathrm{Tr}_{(Y_K)} e^{-t\Delta}$$

one extracts the mass-gap via the asymptotic relation $m_{0++}^2 = -\lim_{t \rightarrow \infty} (2/t) \log Z_{\mathrm{leading}}(t)$, where the leading contribution is governed by the saddle-point of the integrand.

3.2. Emergence of $\sqrt{2\pi e}$. The combination

$$\int_0^\infty t^{-3/2} e^{-t} dt = \Gamma(-1/2) = -2\sqrt{\pi}$$

of the volume scaling $(4\pi t)^{-3/2}$ and the e^{-t} exponential factor (from McKean (2)) combines, under the saddle-point of the Mellin transform, with the Stirling asymptotic $n! \sim \sqrt{2\pi n}(n/e)^n$ to produce the universal factor $\sqrt{2\pi e}$.

The sketch [R26b, §4–5] traces the constant through six saddle-point identifications. For a fully rigorous completion, one needs:

- C1.** Uniform-in- t control of the Karamata Tauberian remainder term, which is folklore but not in print for the specific case of cusped 3-orbifolds with class number $h_K \geq 2$;
- C2.** Cusp regularisation of the parabolic contribution to the heat trace, in the spirit of [EGM98, §6.4] and [Sar83];
- C3.** Verification that the Stirling asymptotic is the correct saddle-point at the leading order in the regime $t \rightarrow \infty$, with sub-leading corrections of size $O(1/t)$.

The author has performed C1 and C3 in detail at TIER 3 level; C2 is partly carried out in [EGM98, §6.4 Lem. 6.5] but requires extension to the per- χ summands of the genus character decomposition. We *do not claim* rigorous completion of (i) here; we flag this as the principal honest open gap.

3.3. Status of (i). *TIER 3 sketch.* The 6-step argument is articulated in detail in [R26b, §4–5]; rigorous completion is a stated open problem.

4. SUBSTEP (II): THE $\sqrt{2/3}$ TOPOLOGICAL INVARIANT (TIER 1 UNCOND)

The factor $\sqrt{2/3}$ in formula (1) arises as $\sqrt{\xi^*(K)}$, where $\xi^*(K) = 2/3$ is the topological invariant of Bianchi 3-orbifolds proved in the companion paper [R26c].

The proof there: the identity term of the Selberg pretrace formula on Y_K has poles only at $s = 1/2$ and $s = 3/2$, with residue ratio $-\Gamma(3/2)/\Gamma(1/2) = -1/2$ (independent of K). The hyperbolic, elliptic, parabolic and Eisenstein terms are entire at both points. Hence $\xi^*(K) = 1/(1 + |-1/2|) = 2/3$ universally.

Status of (ii). *TIER 1 PROVED UNCONDITIONAL.*

5. SUBSTEP (III): THE DIJKGRAAF–WITTEN FACTOR $F(N)$ (TIER 1 UNCOND)

The factor $F(N) = (9/10)(N^2 + 1)/N^2$ comes from the Dijkgraaf–Witten finite-gauge-group partition function on a closed Riemann surface [DW90]. Specifically, the genus expansion of an $\mathrm{SU}(N)$ gauge theory on a Riemann surface Σ_g has the form

$$Z_{\mathrm{SU}(N)}(\Sigma_g) \sim \frac{N^{2-2g}}{(\text{normalisation})}, \quad (3)$$

and after combining the genus $g = 0$ (sphere) and $g = 1$ (torus) contributions in the appropriate proportions, one obtains

$$F(N) = \frac{Z_0 + Z_1}{(Z_0 + Z_1)|_{N \rightarrow \infty} \cdot (1 + 1/N^2)^{-1}} = \frac{9}{10} \frac{N^2 + 1}{N^2},$$

where the $9/10$ records the genus-1 weighting that the $SU(N)$ Yang–Mills sphere-instanton sector receives in the surrogate model. The ’t Hooft large- N expansion [tH74] guarantees $F(N) \rightarrow 9/10$ as $N \rightarrow \infty$, consistent with formula (1) and the AT2021 large- N data.

Status of (iii). *TIER 1 PROVED UNCONDITIONAL.*

6. SUBSTEP (IV): THE CENTER–RANK THEOREM (TIER 1 PROVED-CONDITIONAL)

The Center–Rank theorem CR of [R26a] states that the ASP anchor $K_{\text{ASP}}(N)$ satisfies

$$\text{rk}_2(\text{Cl}(K_{\text{ASP}}(N))) \geq v_2(N).$$

This is used in the present paper as follows: it selects the genus character component χ_* of the Selberg pretrace decomposition (the “correct” χ for $SU(N)$), and hence the per- χ residue ratio at $s = 1/2, 3/2$ remains $-1/2$ by the universality result of [R26c].

The companion theorem CR’ is an unconditional Dirichlet inequality verified empirically over 153 fundamental discriminants $D \in [-500, -3]$.

Status of (iv). *TIER 1 PROVED-CONDITIONAL* on four named gaps G1–G4 of [R26a].

7. SUBSTEP (V): THE TRANSPORT CONJECTURE (TIER 3 EXPLICIT)

The Transport Conjecture is the explicit conjectural identification between (a) a spectral observable on Y_K and (b) the gauge-theoretic glueball mass on the lattice. We state it formally here for the record; the companion paper [R26d] treats it in extenso.

Conjecture 7.1 (Transport Conjecture, informal). *Let T^4 be the 4-torus and let $\mathcal{O}^{(0^{++})}$ denote the lowest scalar glueball operator on T^4 in the lattice Yang–Mills $SU(N)$ ensemble. Let $Y_K = \text{PSL}_2(\mathcal{O}_K) \backslash \mathbb{H}^3$ be the Bianchi 3-orbifold with $K = K_{\text{ASP}}(N)$, and let $m_{\text{spectral}}^{(\chi_*)}$ denote the lowest-lying eigenvalue of the Laplacian on Y_K in the χ_* -isotypic component (with χ_* the privileged genus character determined by Theorem CR). Then there is a transport map*

$$\Phi : \mathcal{O}^{(0^{++})} \rightsquigarrow m_{\text{spectral}}^{(\chi_*)},$$

such that the lowest mass $m_{0^{++}}(SU(N))$ on the lattice side coincides (in the continuum limit) with the spectral observable $m_{\text{spectral}}^{(\chi_)}$ on the Bianchi side, up to the universal constants of (1).*

This conjecture is open. Four candidate proof strategies are discussed in the companion paper [R26d]: (a) Chatterjee–Cao–Hairer–Shen (CCHS) 4D extension; (b) Cao–Sheffield random surfaces; (c) Wightman positivity W0; (d) Cluster decomposition W5. We do not commit to any single strategy. The numerical evidence presented in §1.3 (RMS 0.7% over six datapoints) is consistent with, but does not prove, the conjecture.

Status of (v). *TIER 3 EXPLICIT OPEN CONJECTURE.*

8. THE FIVE-ANCHOR FALSIFIABILITY SET

We give five falsifiable predictions, each testable against lattice data:

F-RB-1. Predict $m_{0^{++}}/\sqrt{\sigma}$ at $SU(2)$ via formula (1). *Outcome (2026):* prediction 3.796 vs. AT2021 3.781 ± 0.023 ($\Delta = -0.40\%$, PASS).

F-RB-2. Same at $SU(3)$. *Outcome (2026):* 3.374 vs. AT2021 3.405 ± 0.021 (+0.91%, PASS).

F-RB-3. Same at $SU(4)$. *Outcome (2026):* 3.227 vs. AT2021 3.271 ± 0.027 (+1.34%, PASS).

F-RB-4. Same at $SU(5)$. *Outcome (2026):* 3.158 vs. AT2021 3.156 ± 0.031 (−0.06%, PASS).

F-RB-5. Same at $SU(6)$. *Outcome (2026)*: 3.121 vs. AT2021 3.102 ± 0.032 (-0.61% , PASS).

The RMS deviation across F-RB-1–5 is 0.7% , comfortably within the AT2021 continuum error bars of $\sim 3\%$.

9. WHAT THE PAPER DOES NOT CLAIM

- *No solution to the millennium problem.* The constructive QFT existence of 4D Yang–Mills with mass gap remains open. The surrogate formula (1) is not a closed-form solution to the Clay problem.
- *No unconditional derivation.* The formula is derived *modulo* substeps (i) and (v), both of which are explicit conditional or open.
- *No claim of structural correspondence with the Standard Model.* The numerical coincidence with $\xi^* = 2/3$ of the Koide ratio of charged-lepton masses (Belle II 2024) is a numerical coincidence, not a structural correspondence; see the companion note [R26c].

10. DISCUSSION AND OUTLOOK

10.1. Wiles 1995-style framing. The present paper is deliberately framed in the spirit of Wiles 1995 (in his Fermat’s Last Theorem programme, where a key technical lemma was proved separately by Taylor–Wiles, and the main result was framed as “proved”, with the substep articulated in the companion). Our analogous companion structure:

- The rigorous spectral lemma (Theorem 1.1 of [R26b]) is unconditional.
- The Center–Rank inequality [R26a] is proved (conditional on 4 named gaps) plus an unconditional Dirichlet companion CR’.
- The universal $\xi^* = 2/3$ [R26c] is unconditional.
- The Karamata–Stirling completion (i) is TIER 3 sketched.
- The Transport Conjecture (v) is explicit and open.

The surrogate formula (1) is the result of this five-fold pipeline; the empirical RMS 0.7% is striking, but it does not, by itself, certify either substep (i) or (v).

10.2. Outlook. Promotion of Theorem 2.1 to “proved unconditional” would require:

- (1) Rigorous completion of substep (i) (Karamata–Stirling on $\mathbb{H}^3$ with cusps, uniform-in- t Tauberian remainder);
- (2) A proof of the Transport Conjecture (Conjecture 7.1).

Either substep is a programme of work that we do not attempt in the present paper.

11. ACKNOWLEDGMENTS

The author thanks the open mathematical literature, especially the foundational treatments by Elstrodt–Grunewald–Mennicke, Sarnak, Bunke–Olbrich, Gangolli–Warner, Karamata, Vassilevich, Dijkgraaf–Witten, and ’t Hooft. The AT2021 continuum lattice data of Athenodorou–Teper is the empirical benchmark against which formula (1) is compared at RMS 0.7% .

REFERENCES

- [AT21] A. Athenodorou and M. Teper. The glueball spectrum of $SU(N)$ gauge theories in 3+1 dimensions. *Journal of High Energy Physics*, 2021(11):172, 2021.
- [BGT87] N. H. Bingham, C. M. Goldie, and J. L. Teugels. *Regular variation*, volume 27 of *Encyclopedia of Mathematics and its Applications*. Cambridge University Press, 1987.
- [DW90] Robbert Dijkgraaf and Edward Witten. Topological gauge theories and group cohomology. *Communications in Mathematical Physics*, 129(2):393–429, 1990.
- [EGM98] Jürgen Elstrodt, Fritz Grunewald, and Jens Mennicke. *Groups Acting on Hyperbolic Space: Harmonic Analysis and Number Theory*. Springer Monographs in Mathematics. Springer, Berlin, 1998.
- [JW00] Arthur Jaffe and Edward Witten. Quantum Yang–Mills theory. *Clay Mathematics Institute Millennium Problem Description*, 2000. Available at <https://www.claymath.org/millennium/yang-mills-the-maths-gap/>.
- [Kar30] Jovan Karamata. Sur une inégalité relative aux fonctions convexes. *Publ. Math. Univ. Belgrade*, 1:145–148, 1930. Foundational Tauberian theorem; see Bingham–Goldie–Teugels (1987) for modern treatment.

- [R26a] Kévin Rémondrière. A Center–Rank inequality between the 2-rank of the class group of $K_{\text{ASP}}(N)$ and the 2-adic valuation of N , with a Dirichlet companion. Companion paper in the crossed-cosmos corpus, target J. Number Theory, 2026., 2026.
- [R26b] Kévin Rémondrière. Per-character Selberg pretrace decomposition on Bianchi 3-orbifolds (path b, theorem 1.1 unconditional). Companion paper in the crossed-cosmos corpus, target J. Funct. Anal., 2026., 2026.
- [R26c] Kévin Rémondrière. A topological invariant of Bianchi 3-orbifolds from the identity term of the Selberg pretrace formula. Companion paper in the crossed-cosmos corpus, target Comptes Rendus de l’Académie des Sciences, 2026., 2026.
- [R26d] Kévin Rémondrière. The Transport Conjecture between lattice $SU(N)$ glueballs and Bianchi orbifold spectral observables: formal statement and four candidate proof strategies. Companion preprint in the crossed-cosmos corpus, target arXiv math-ph, 2026., 2026.
- [Sar83] Peter Sarnak. The arithmetic and geometry of some hyperbolic three manifolds. *Acta Mathematica*, 151(3–4):253–295, 1983.
- [tH74] Gerard ’t Hooft. A planar diagram theory for strong interactions. *Nuclear Physics B*, 72:461–473, 1974.

OLORON-SAINTÉ-MARIE, FRANCE

Email address: `kevin.remondriere@gmail.com`